A Machine Learning Analysis of Commercialization Within the #tcm Creator Economy

Victoria Tan

This manuscript was compiled on June 7, 2026

The project utilizes data from Facebook that contain #tcm and pulls data from the past year. The purpose of the project is to analyze post caption data to determine the changing use of #tcm by various types of accounts. While the acronym tcm has various other meanings, this project specifically analyzes tcm as Traditional Chinese Medicine. Through this analysis, the project tracks how independent creators navigate commercialization and professionalization over time. The results reveal that commercialization is a non-linear process. The model demonstrates that rather than over commercializing indefinitely, creators in the #tcm space strategically limit explicit business signals to protect audience trust and their social capital.

tcm | Commercialization | Creator Economy |

Significance

This project is critical in understanding how digital subcultures navigate the varying internal and external pressures of cultural appropriation and commercialization. By tracking #tcm (Traditional Chinese Medicine), it is possible to analyze the viral ”Chinamaxxing” trend through the lens of one of its core subcultures. The machine learning evaluation pipeline exposes the critical flaws of training a model for accuracy rather than precision, demonstrating the severe implications this technical limitation has on erasing entire minority communities due to major class imbalances in the data. Overall, this project demonstrates a structural stabilization within the creator economy regarding the use of explicit commercial language and explores the sociological reasons driving this trend.

1. Introduction

At the start of 2026, the total global number of active social media identities hit just over 5.5 billion users. This massive scale of digital participation directly contributes to the rapid growth of cultural exchange and connectivity across the world. Platforms like Instagram and TikTok have made experiencing and learning about global cultures highly accessible. Over the past year, 'Chinamaxxing' has become an increasingly popular trend where Western creators explore Chinese cultural practices(1). A primary example of a hashtag that falls under this umbrella is #tcm, which stands for Traditional Chinese Medicine. This particular hashtag tracks trends showing creators adopting holistic health practices or wellness traditions.

Social media platforms have brought niche parts of various cultures to the forefront of mainstream media. Through these digital channels, Western and Asian creators have an equal opportunity to freely share or explore Eastern cultural traditions (2), allowing Asian creators to thrive by sharing their personal experiences. Despite creators' ability to develop independent content, they inevitably fall into a structural trap that forces them to use specific language that performs well within the bounds of a platform's data infrastructure (3). Evaluating this data infrastructure reveals a rigid binary of personal curation versus commercial intent hidden directly within post caption data.

During analysis of social media text data, common machine learning pipelines consistently default to maximize overall accuracy. Imbalanced data is particularly vulnerable as these pipelines create a mathematical shortcut which inevitably erases niche subcultures because of a preference for dominant trends. The analysis of social media text data using a machine learning model links the sociological concept of structural invisibility with the shortcomings of measurable data science evaluations. Additionally, this analysis demonstrates a new framework that optimizes for precision rather than overall accuracy, which is a starting point for reevaluating these limitations of these existing classification algorithms.

2. Source and Collection

The data for this study was collected from Facebook, capturing posts that utilized the #tcm hashtag over a one-year period between May 13, 2025, and May 13, 2026. Upon initial inspection, the fields structuring the dataset included creation time, language, post owner ID, post owner type, and the raw caption text. Among these fields, the account type classification (post_owner.type) was particularly useful because it categorized users into three distinct groups: businesses, creators, and personal accounts. A total of 7,588 unique creator accounts were isolated for the primary text corpus, providing a robust baseline to separate independent creator activity from corporate and purely personal profiles.

Initially, the raw data was split into two separate files, each containing six months worth of post data. After concatenating the dataframes chronologically, the data was filtered to isolate English-language posts. The raw caption data was then processed using regular expressions and natural language tokenization to separate the hashtags from the core caption text. This preprocessing step created two new structural columns: hashtags_extracted and text_clean, isolating either the explicit hashtag strings or the clean text blocks for downstream feature engineering.

3. Methods

A. Data.

0.0.1. Initial visualizations. After the data was preprocessed, initial visualizations tracking the volume and proportion of posts across different account types were created to better understand our data landscape. Observing the raw counts of each account type demonstrated that there were 9,290 posts made by creators, 6,132 made by business accounts, and 186 from personal accounts. Examining these raw counts justified isolating creator accounts for the primary text corpus; business accounts would clearly possess commercial intent from the outset, while personal accounts were simply too few in number to appropriately utilize. Furthermore, the timeline visualizations depicting the volume and proportion of #tcm posts exposed a significant class difference. As seen in Figure 1 and Figure 2, there is a significantly higher proportion of creator accounts relative to business and personal profiles over time. Additional parts of the data preprocessing portion highlighted other common hashtags frequently paired with #tcm. While tags like #holistichealth or #wellness clearly demonstrate a specific subculture of Chinese health practices, the analysis also revealed that the abbreviation #tcm can be interpreted in entirely separate ways—such as #tcmnews, which represents a news station based in Pakistan.

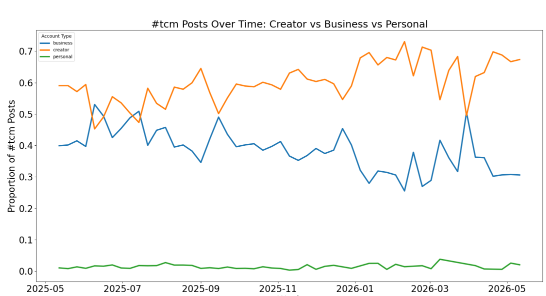

Fig. 1. Post proportions over time across account types.

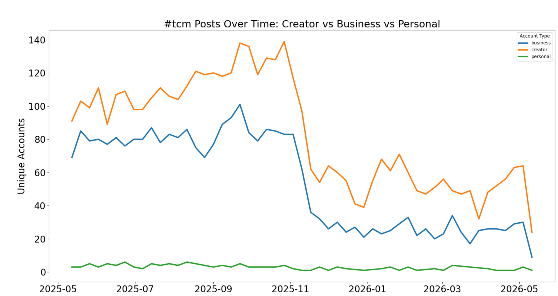

Fig. 2. Unique user distributions tracking independent engagement.

B. Statistical Analysis.

0.0.2. Logistic Regression. The statistical analysis of the data was conducted using an L1-regularized logistic regression model. This model was chosen because it is capable of classifying creator captions based on user intent. The target labels—personal curation (0) or commercial optimization (1)—were identified by employing a heuristic regular expression function to determine if a specific caption contained commercial intent. This regex function targeted hyper-standardized monetization indicators and phrases within the caption text, such as 'promo code', 'buy now', and 'shop now'. To test the framework's predictive power, the model's evaluation relied on tracking raw accuracy, precision, and recall. This approach was chosen to demonstrate the limitations of relying strictly on raw accuracy, illustrating how it is insufficient when working with data that exhibits a significant class imbalance.

0.0.3. Visualizations. Following the logistic regression model, three visualizations were created to analyze the performance and predictions of the framework over time. The first, Figure 3 (3), tracks the percentage of creator captions that contain commercial signals mapped across our timeline. It highlights a peak of over 18% in mid-2025, followed by a significant downward shift to roughly 6% in early 2026. The second visualization, Figure 4 (4), utilizes an absolute stackplot to display the raw total of posts over time, broken down into personal captions (0) and business/commercial captions (1) to show the baseline volume of the dataset. The third visualization, Figure 5 (5), offers a macro-comparison by year to evaluate the shifting concentration of commercial intent. This final chart confirms a 12.3% average of business content on creator accounts during the 2025 window compared to a lower baseline of 8.3% in 2026.

0.1. Statistical Analysis.

0.1.1. Logistic Regression. The statistical analysis of the data was conducted using an L1-regularized logistic regression model. This model was chosen because it is capable of classifying creator captions based on user intent. The target labels—personal curation (0) or commercial optimization (1)—were identified by employing a heuristic regular expression function to determine if a specific caption contained commercial intent. This regex function targeted hyper-standardized monetization indicators and phrases within the caption text, such as "promo code," "buy now," and "shop now." To test the framework's predictive power, the model's evaluation relied on tracking raw accuracy, precision, and recall. This approach was chosen to demonstrate the limitations of relying strictly on raw accuracy, illustrating how it is insufficient when working with data that exhibits a significant class imbalance.

0.1.2. Visualizations. Following the logistic regression model, three visualizations were created to analyze the performance and predictions of the framework over time (see Figure 6). The first, Panel (a) (3), tracks the percentage of creator captions containing commercial signals across our timeline, highlighting a peak of over 18% in mid-2025 followed by a significant downward shift to roughly 6% in early 2026. The second visualization, Panel (b) (4), utilizes an absolute stackplot to display the raw total of posts over time broken down into personal and commercial categories, providing empirical proof of the severe class imbalance and demonstrating the massive footprint of organic posts the model must navigate. The third visualization, Panel (c) (5), offers a macro-comparison by year to evaluate the shifting concentration of commercial intent, confirming a 12.3% average of business content on creator accounts during the 2025 window compared to a lower baseline of 8.3% in 2026.

0.1.3. Visualizations. Following the logistic regression model, three visualizations were created to analyze the performance and predictions of the framework over time. An important finding highlighted by these figures was the significant class imbalance in the data; when applied to a typical machine learning standard of training for overall accuracy, it inevitably erases a subcategory of participants within the data. This major class imbalance is illustrated in Figure 4, where a significant number of personal posts (0) dominate over commercial posts (1). More specifically, the model identified that 88.6% of the data belong to class 0, or personal captions. Without tuning the model appropriately, it would have achieved a high overall accuracy score but also a recall score of 0%, meaning that it would have failed to detect any commercial posts. However, training the model toward optimizing precision and recall made it possible for the model to achieve a high classification integrity. When tuning the model, it was discovered that a C-value of 10.0 achieved the highest precision score of 93.08%.

4. Visualizations

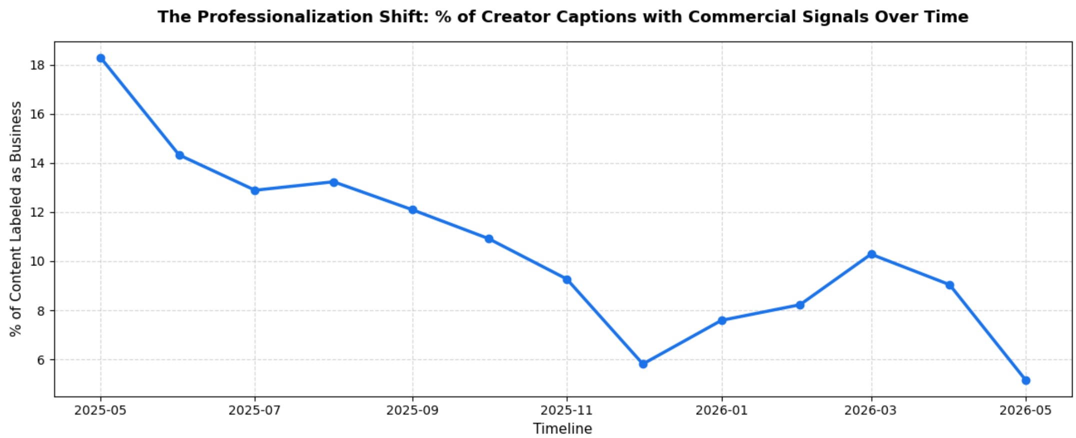

Fig. 3. The Professionalization Shift

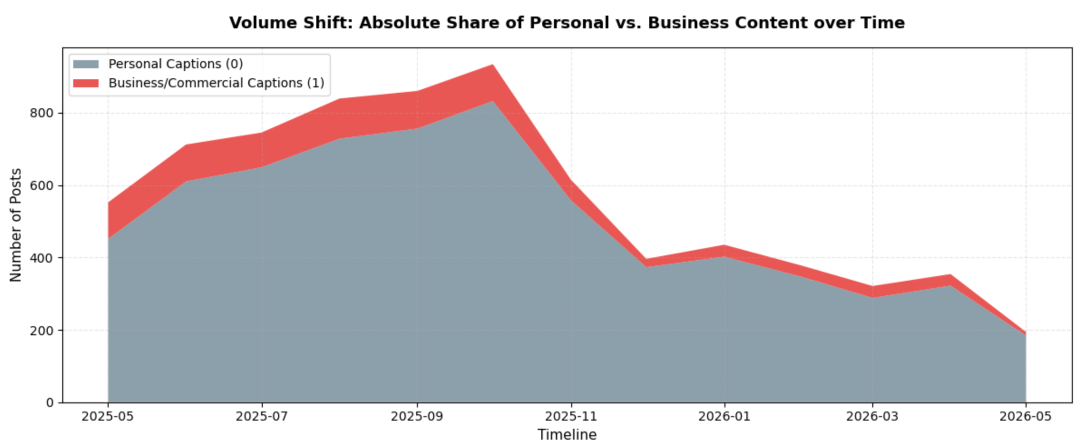

Fig. 4. Volume Shift Stackplot

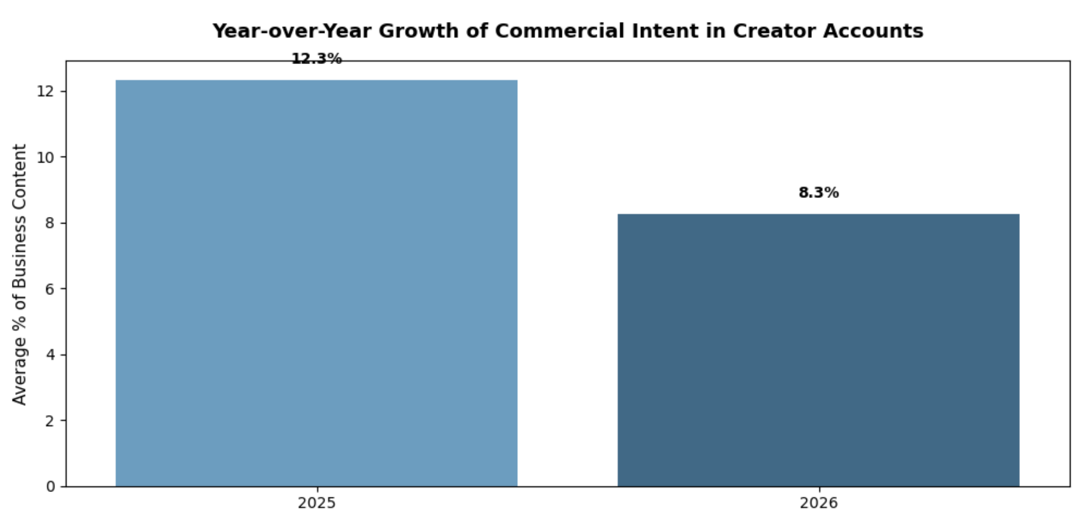

Fig. 5. Year-over-Year Growth Comparison

Fig. 6. Longitudinal analysis of creator captions following logistic regression classification, tracking metric trends over time (a), absolute volume distributions (b), and macro shifts across calendar years (c).

5. Results

An important finding about the model was the significant class imbalance in the data; when applied to a typical machine learning standard of training for overall accuracy, it inevitably erases a subcategory of participants within the data. This major class imbalance is illustrated in 4, where a significant number of personal posts (0) dominate over commercial posts (1). More specifically, the model identified that 88.6% of the data belong to class 0, or personal captions. Without tuning the model appropriately, it would have achieved a high overall accuracy score but also a recall score of 0%, meaning that it would have failed to detect any commercial posts. However, training the model toward optimizing precision and recall made it possible for the model to achieve a high classification integrity. When tuning the model, it was discovered that a C-value of 10.0 achieved the highest precision score of 93.08%.

Another key finding produced by the model was the overall downward shift in commercial intent among creator accounts. This shift is apparent after aggregating over the calendar year and utilizing the average concentration of commercial intent in posts, which showed a 4% decrease from 2025 to 2026 (refer to 5). This trajectory is also visible in 3, which tracks a downward trend in commercial intent that began to stabilize around the start of 2026. The stabilization of commercial intent over time can reflect a behavioral shift in the ways that creators approach commercialization.

6. Discussion

Given the results derived from the model, we can conclude two key points of sociological discussion. The first is the maturation of the creator economy, which is illustrated through the structural stabilization of commercial signals—specifically, the yearly average dropping from 12.3% in 2025 to 8.3% in 2026. This maturity within the digital economy can be attributed to a conscious effort by creators to avoid oversaturating their accounts with obvious commercial content. Instead, creators limit their explicit use of business signals to preserve audience trust and safeguard their social capital ((2)). This strategic restraint is particularly evident among creators who build their professional images around niche, culture-driven subcultures, such as those engaging with the ”Chinamaxxing” trend and utilizing hashtags like #tcm to channel engagement. The second point of discussion centers on how this restraint actively protects the social capital of these subcommunities. By refraining from aggressive commercial buzzwords, creators maintain a baseline of authenticity that shields their audiences from overly corporate jargon and the intrusive influence of mass commercial spaces as a whole. As this relates to #tcm, it demonstrates a creator's power to preserve intimate community bonds, ensuring that corporate professionalization does not erode the collective solidarity of the subculture.

Additionally, the performance of the model identifies a fundamental flaw in machine learning evaluation solely through text-based caption analysis. This flaw overlooks the rapid adaptation of algorithmic biases on the part of creators, who best understand the buzzwords that draw attention to their content. For instance, while a classifier can be trained to target explicit marketing phrases like ”promo code” or ”buy now,” a creator's understanding of these platform mechanisms exposes opportunities to veer away from these buzzwords and pivot toward implicit commercialization. By doing so, they are able to preserve their audience and avoid this algorithmic bias by instead utilizing embedded brand tagging or visual product placements.This limitation is further exacerbated by the fact that text-based analysis is insufficiently equipped to keep up with the rapidly changing keywords and slangs associated with social commerce. Within the context of the #tcm subculture, this analytical boundary directly overlaps with deeper socio-emotional realities; specifically, hyper-visibility and passive content consumption are frequently associated with heightened traumatic stress among minorities in online spaces ((3)). Therefore, the observed decline in commercial signals within the data might not exclusively reflect economic maturity; instead, it could also be attributed to a voluntary separation from commercialization, where creators deliberately choose to stop participating publicly.

In the future, it is possible to circumvent the limitations discussed by expanding the scope of evaluation beyond just text classification. This could be done by pairing text classification evaluations with visual tracking of embedded logos or tags, creating a more fitting landscape of data reflective of modern social platforms. In addition, these approaches could pair with methods that evaluate the attention-seeking behavior of various subcultures to better inform statistical analysis later. The goal of this data analysis project was to understand how creators navigate commercialization and professional advances on social media platforms. Through this work, we have been able to understand that the commercialization of creator accounts is a non-linear process. During the tuning process of the model, it was possible to identify that subcultures like #tcm were able to successfully negotiate a baseline ratio for personal to commercial intended posts. This system is particularly important for the survival of this subculture because it protects viewers from an overcommercialized community and maintains trust between creators and their viewers. However, this equilibrium is increasingly complicated by Western influencers adopting the #tcm hashtag and its underlying ”Chinamaxxing” themes. While these Western creators utilize the subculture's specific aesthetic as a strategic choice to accumulate engagement capital, their entry introduces a layer of commercial pressure that threatens to cause a context collapse within the space. This crossover forces the core minority subculture to reinvent its linguistic boundaries or retreat deeper into implicit visual codes to keep the community safe from external exploitation. Overall, our model demonstrates that despite these encroaching cross-cultural and corporate dynamics, independent creators have successfully established a resilient, mature baseline ecosystem that dynamically balances professionalization with authentic digital solidarity.

1. CE Noticias Financieras English, 'chinamaxxing', the social media trend between posturing and curiosity about the asian giant. *CE Noticias Financieras, Engl. ed.* (2026) ProQuest Document ID: 3306213127.
2. EY Jung, Transnational migrations and youtube sensations: Korean americans, popular music, and social media. *Ethnomusicology* **58**, 54–82 (2014).
3. Cc Yang, JY Tsai, Asians and asian americans' social media use for coping with discrimination: A mixed-methods study of well-being implications. *Heliyon* **9**, e16842 (2023) Corrigendum published in Heliyon, 10(12), 2024, e32471.
4. M Cho, Anonymous, qanon, tik-tok teens, k-pop fans. *NECSUS. Eur. J. Media Stud.* **11**, 169–193 (2022).
5. ZF Chen, R Sun, W Tao, Channeling engagement into action: the role of empowerment in asian americans' social media use in combating anti-asian discrimination. *Asian J. Commun.* **34**, 236–257 (2024).
6. K Ng, What race does online: "gangnam style" and asian/american identity in the digital age. *Verge: Stud. Glob. Asias* **1**, 109–113 (2015).
7. H Dhillon, The making of modern us citizenship and alienage: The history of asian immigration, racial capital, and us law. *Law Hist. Rev.* **41**, 1–42 (2023).
8. CJ Kim, The racial triangulation of asian americans. *Polit. & Soc.* **27**, 105–138 (1999).
9. J Xu, JC Lee, The marginalized 'model' minority: An empirical examination of the racial triangulation of asian americans. *Soc. Forces* **91**, 1363–1397 (2013).

7. Agentic Analysis

Throughout the statistical analysis portion of this project, I utilized Claude to better understand the machine learning model and the logistic regression code from class, adapting that workflow to fit my own project requirements by creating a custom evaluation criteria with regular expressions to identify commercial phrases. When I ran into RAM limitations on my laptop while running the regression, I cleared background applications and used the AI to help optimize code and reduce the memory overhead. For the website development, I followed the course demo to ”vibe code” using Claude, but ran out of tokens mid-session and was left with incomplete scripts. To get around this roadblock, I pivoted to Gemini to generate the initial foundational code for the website, accepting its layout skeleton while rejecting its over-complicated styling suggestions, and manually edited the remaining details in PyCharm to refine the final product.